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ASSOCIATE SERIES · A02 · SECTION 2 INGESTION (21%) QUICK REFERENCE

Batch · incremental · streaming

Section 2: ingestion patterns (batch, streaming, incremental), local files, Lakeflow Connect connectors. Free Edition: serverless, Unity Catalog volumes, Auto Loader. Fictional orders only.

1. Three patterns

```
batch        spark.read ... load(src)        reads EVERY file, every run
incremental  Auto Loader + checkpoint         only files not yet processed
streaming    same engine, runs continuously   the trigger decides how often
```

Batch + append = duplicates on every rerun. Batch should overwrite.

2. Landing zone: a Unity Catalog volume

```
CREATE VOLUME workspace.bt_ingest_xxxxxxxxxxxx.landing;
-- files: /Volumes/workspace/bt_ingest_xxxxxxxxxxxx/landing/orders/
dbutils.fs.put(f"{src}/orders_day1.csv", day1)
```

A volume stores files under Unity Catalog governance.

3. Batch load (comma-separated values, CSV)

```
df = (spark.read.format("csv").option("header", "true")
        .schema("order_id INT, customer STRING, amount DOUBLE, order_date DATE").load(src))
df.write.mode("overwrite").saveAsTable("...orders_batch")
```

Day 1: 3 rows. Day 2: re-reads both files, 5 rows.

4. Incremental load: Auto Loader

```
(spark.readStream.format("cloudFiles").option("cloudFiles.format", "csv")
   .option("header", "true").schema(order_schema).load(src)
   .withColumn("source_file", col("_metadata.file_name"))
 .writeStream.option("checkpointLocation", ckpt)
   .trigger(availableNow=True).toTable("...orders_bronze"))
```

Day 2 reads only orders_day2.csv (2 rows). The checkpoint loads each file once.

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Triggers, local files, Lakeflow Connect

5. Triggers on serverless

```
availableNow=True        process all new data, then stop   (serverless OK)
once=True                deprecated, still supported
processingTime / continuous   NOT on serverless notebooks or jobs
                         -> INFINITE_STREAMING_TRIGGER_NOT_SUPPORTED
```

Always-on: a pipeline in continuous mode, or availableNow in a continuous job. Schedule incremental runs with Lakeflow Jobs.

6. Local files

```
New > Add or upload data > Create or modify table   one-time load -> managed Delta table
Catalog > volume > Upload to this volume           keep files, read them with code
```

Uploads suit small reference files, not data that keeps arriving.

7. Lakeflow Connect connectors

```
standard  Spark + Structured Query Language (SQL) application programming interfaces (APIs)
          Auto Loader, COPY INTO, Structured Streaming (Kafka). You maintain the logic.
managed   connection + ingestion pipeline, configured not coded. Software as a service
          (SaaS) like Salesforce, Workday, SharePoint, or SQL Server databases.
```

Managed: authentication, retries, schema changes handled. Free Edition: needs a real source; database change data capture (CDC) is paid-only. Concepts here.

8. Scenario words

Small, fully replaced, daily snapshot, one-time upload -> batch (overwrite). New files keep arriving, process only new data, never reprocess -> Auto Loader + checkpoint. Seconds or minutes latency -> streaming. Supported SaaS app or database -> managed connector.

Practice challenge

Day 3 adds 4 orders; run both loads. Answer: orders_batch 9 (overwrite); Auto Loader reads 4, bronze 9; if all three batch runs appended: 3 + 5 + 9 = 17 rows with duplicates.

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